

Replication of “null results” – Absence of evidence or evidence of absence?

Samuel Pawel^{1*†}, Rachel Heyard^{1†}, Charlotte Micheloud¹, Leonhard Held¹

*For correspondence:
samuel.pawel@uzh.ch (SP)

†Contributed equally

¹Epidemiology, Biostatistics and Prevention Institute, Center for Reproducible Science, University of Zurich, Switzerland

Abstract In several large-scale replication projects, statistically non-significant results in both the original and the replication study have been interpreted as a “replication success”. Here we discuss the logical problems with this approach: Non-significance in both studies does not ensure that the studies provide evidence for the absence of an effect and “replication success” can virtually always be achieved if the sample sizes are small enough. In addition, the relevant error rates are not controlled. We show how methods, such as equivalence testing and Bayes factors, can be used to adequately quantify the evidence for the absence of an effect and how they can be applied in the replication setting. Using data from the Reproducibility Project: Cancer Biology, the Experimental Philosophy Replicability Project, and the Reproducibility Project: Psychology we illustrate that many original and replication studies with “null results” are in fact inconclusive. We conclude that it is important to also replicate studies with statistically non-significant results, but that they should be designed, analyzed, and interpreted appropriately.

Introduction

Absence of evidence is not evidence of absence – the title of the 1995 paper by Douglas Altman and Martin Bland has since become a mantra in the statistical and medical literature (Altman and Bland, 1995). Yet, the misconception that a statistically non-significant result indicates evidence for the absence of an effect is unfortunately still widespread (Greenland, 2011; Makin and de Xivry, 2019). Such a “null result” – typically characterized by a p -value $p > 0.05$ for the null hypothesis of an absent effect – may also occur if an effect is actually present. For example, if the sample size of a study is chosen to detect an assumed effect with a power of 80%, null results will incorrectly occur 20% of the time when the assumed effect is actually present. If the power of the study is lower, null results will occur more often. In general, the lower the power of a study, the greater the ambiguity of a null result. To put a null result in context, it is therefore critical to know whether the study was adequately powered and under what assumed effect the power was calculated (Hoenig and Heisey, 2001; Greenland, 2012). However, if the goal of a study is to explicitly quantify the evidence for the absence of an effect, more appropriate methods designed for this task, such as equivalence testing (Wellek, 2010; Lakens, 2017; Senn, 2021) or Bayes factors (Kass and Raftery, 1995; Goodman, 1999, 2005; Dienes, 2014; Keyser et al., 2020), should be used from the outset.

The interpretation of null results becomes even more complicated in the setting of replication studies. In a replication study, researchers attempt to repeat an original study as closely as possible in order to assess whether consistent results can be obtained with new data (National Academies of Sciences, Engineering, and Medicine, 2019). In the last decade, various large-scale replication

projects have been conducted in diverse fields, from the biomedical to the social sciences (*Prinz et al., 2011; Begley and Ellis, 2012; Klein et al., 2014; Open Science Collaboration, 2015; Camerer et al., 2016, 2018; Klein et al., 2018; Cova et al., 2018; Errington et al., 2021*, among others). The majority of these projects reported alarmingly low replicability rates across a broad spectrum of criteria for quantifying replicability. While most of these projects restricted their focus on original studies with statistically significant results (“positive results”), the *Reproducibility Project: Cancer Biology* (RPCB, *Errington et al., 2021*), the *Experimental Philosophy Replicability Project* (EPRP, *Cova et al., 2018*), and the *Reproducibility Project: Psychology* (RPP, *Open Science Collaboration, 2015*) also attempted to replicate some original studies with null results – either non-significant or interpreted as showing no evidence for a meaningful effect by the original authors.

Although the EPRP and RPP interpreted non-significant results in both original and replication study as a “replication success” for some individual replications (see, for example, the replication of *McCann (2005*, replication report: <https://osf.io/wcm7n>) or the replication of *Ranganath and Nosek (2008*, replication report: <https://osf.io/9xt25>)), they excluded the original null results in the calculation of an overall replicability rate based on significance. In contrast, the RPCB explicitly defined null results in both the original and the replication study as a criterion for “replication success”. According to this “non-significance” criterion, 11/15 = 73% replications of original null effects were successful. Four additional criteria were used to provide a more nuanced assessment of replication success for original null results: (i) whether the original effect estimate was included in the 95% confidence interval of the replication effect estimate (success rate 11/15 = 73%), (ii) whether the replication effect estimate was included in the 95% confidence interval of the original effect estimate (success rate 12/15 = 80%), (iii) whether the replication effect estimate was included in the 95% prediction interval based on the original effect estimate (success rate 12/15 = 80%), (iv) and whether the *p*-value obtained from combining the original and replication effect estimate with a meta-analysis was non-significant (success rate 10/15 = 67%). Criteria (i) to (iii) are useful for assessing compatibility in effect estimates between the original and the replication study. Their suitability has been extensively discussed in the literature. The prediction interval criterion (iii) or equivalent criteria (e.g., the *Q*-test) are usually recommended because they account for the uncertainty from both studies and have adequate error rates when the true effect sizes are the same (*Patil et al., 2016; Mathur and VanderWeele, 2020; Schauer and Hedges, 2021*).

While the effect estimate criteria (i) to (iii) can be applied regardless of whether or not the original study was non-significant, the “meta-analytic non-significance” criterion (iv) and the aforementioned non-significance criterion refer specifically to original null results. We believe that there are several logical problems with both, and that it is important to highlight and address them, especially since the non-significance criterion has already been used in three replication projects without much scrutiny. It is crucial to note that it is not our intention to diminish the enormously important contributions of the RPCB, the EPRP, and the RPP, but rather to build on their work and provide recommendations for ongoing and future replication projects (e.g., *Amaral et al., 2019; Murphy et al., 2022*).

The logical problems with the non-significance criterion are as follows: First, if the original study had low statistical power, a non-significant result is highly inconclusive and does not provide evidence for the absence of an effect. It is then unclear what exactly the goal of the replication should be – to replicate the inconclusiveness of the original result? On the other hand, if the original study was adequately powered, a non-significant result may indeed provide some evidence for the absence of an effect when analyzed with appropriate methods, so that the goal of the replication is clearer. However, the criterion by itself does not distinguish between these two cases. Second, with this criterion researchers can virtually always achieve replication success by conducting a replication study with a very small sample size, such that the *p*-value is non-significant and the result is inconclusive. This is because the null hypothesis under which the *p*-value is computed is misaligned with the goal of inference, which is to quantify the evidence for the absence of an effect. Third, the criterion does not control the error of falsely claiming the absence of an effect at

a predetermined rate. This is in contrast to the standard criterion for replication success, which requires significance from both studies (also known as the two-trials rule, see Section 12.2.8 in [Senn, 2021](#)), and ensures that the error of falsely claiming the presence of an effect is controlled at a rate equal to the squared significance level (for example, $5\% \times 5\% = 0.25\%$ for a 5% significance level). The non-significance criterion may be intended to complement the two-trials rule for null results. However, it fails to do so in this respect, which may be required by regulators and funders. These logical problems are equally applicable to the meta-analytic non-significance criterion.

In the following, we present two principled approaches for analyzing replication studies of null results – frequentist equivalence testing and Bayesian hypothesis testing – that can address the limitations of the non-significance criterion. We use the null results replicated in the RPCB, RPP, and EPRP to illustrate the problems of the non-significance criterion and how they can be addressed. We conclude the paper with practical recommendations for analyzing replication studies of original null results, including simple R code for applying the proposed methods.

Null results from the Reproducibility Project: Cancer Biology

Figure 1 shows effect estimates on standardized mean difference (SMD) scale with 95% confidence intervals from two RPCB study pairs. In both study pairs, the original and replication studies are “null results” and therefore meet the non-significance criterion for replication success (the two-sided p -values are greater than 0.05 in both the original and the replication study). The same is true when applying the meta-analytic non-significance criterion (the two-sided p -values of the meta-analyses p_{MA} are greater than 0.05). However, intuition would suggest that the conclusions in the two pairs are very different.

The original study from [Dawson et al. \(2011\)](#) and its replication both show large effect estimates in magnitude, but due to the very small sample sizes, the uncertainty of these estimates is large, too. With such low sample sizes, the results seem inconclusive. In contrast, the effect estimates from [Goetz et al. \(2011\)](#) and its replication are much smaller in magnitude and their uncertainty is also smaller because the studies used larger sample sizes. Intuitively, the results seem to provide more evidence for a zero (or negligibly small) effect. While these two examples show the qualitative difference between absence of evidence and evidence of absence, we will now discuss how the two can be quantitatively distinguished.

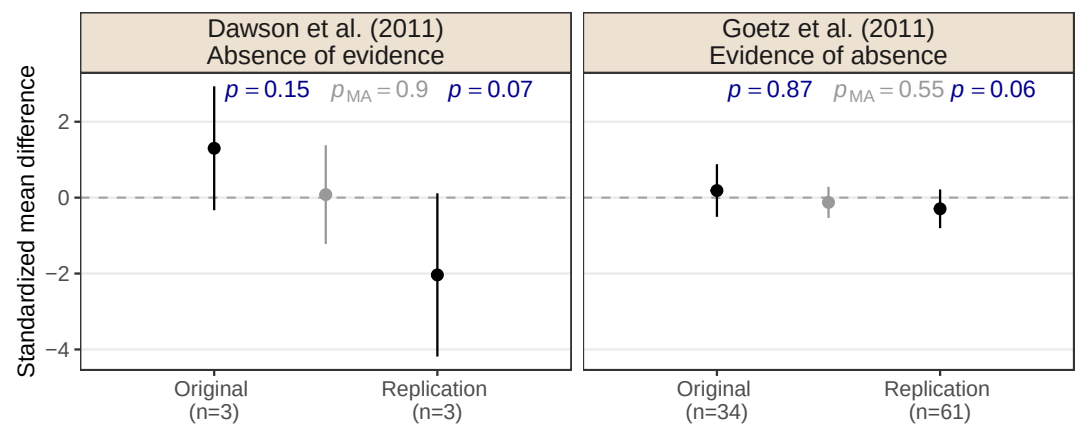


Figure 1. Two examples of original and replication study pairs which meet the non-significance replication success criterion from the Reproducibility Project: Cancer Biology ([Errington et al., 2021](#)). Shown are standardized mean difference effect estimates with 95% confidence intervals, sample sizes n , and two-sided p -values p for the null hypothesis that the effect is absent. Effect estimate, 95% confidence interval, and p -value from a fixed-effect meta-analysis p_{MA} of original and replication study are shown in gray.